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MULTI-DIMENSIONAL FORCE AND TORQUE SENSING ALONG WITH POSE ESTIMATION BASED ON A COMPLIANT FINGERTIP WITH SENSOR-INTEGRATED FLEXIBLE BEAM ARRAYS

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ABSTRACT

Multi-dimensional force and torque sensing and pose estimation are crucial to robotic fingertips to accomplish comprehensive tasks. However, a sensing-integrated compact design to obtain multiaxial forces and torques along with the pose at a high frequency remains challenging. This paper presents a compliant fingertip based on flexible beam arrays, which provide compliance and sensing in the meantime in multiple degrees of freedom. A flexible beam with a curved segment is proposed as the sensory unit for one contact region. A strain gauge sensor based on flexible printed board technology is devised and fabricated, conforming to the curved surface of the beam, to provide deformation information. Sensor readings are sampled through a compact circuit and are transmitted wirelessly to upper computers. On this basis, the compliant fingertip is constructed by 2×2 *flexible beam arrays to perceive the normal and shear forces, bending and twisting torques, and the related poses at the same time. Validation experiments are carried out, showcasing a compact structure and the capability of sensing at over* $1kHz$ *frequency to provide real-time sensing for multi-dimensional force and torque, along with poses. A wine glass arrangement demonstrates its capability as a plug-and-play fingertip for ready integration onto commercial grippers, revealing its potential in multiple applications.*

Keywords: Compliant fingertip; Force and torque sensing; flexible beam array; Integrated sensor; Object manipulation

1. INTRODUCTION

Tactile sensing along with pose estimation are vital for humans to interact with the outer environment in everyday routine, accomplishing object manipulations and perceiving properties, such as shape, position, and roughness. With robotics gradually entering the industrial field and daily applications, requirements for multi-dimensional and real-time tactile sensing are increasing[1]. Compliance and feedback are the two most significant factors that prevent drops and failures for robotic end-effectors, especially dexterous hands and fingertips, which interact with objects directly[2]. The former keeps fragile objects from being crushed, while the latter ensures robust control based on adequate sensing and precise estimation, providing sufficient and preventing excessive contact forces to operate at safe poses.

Enormous efforts have been made to intrinsic tactile sensing, deriving the contact information from sensors that are placed within the structure, especially the joints, of the robot[3–5]. Nevertheless, more delicate operations demand more abundant information involving multiaxial forces and torques. To deal with this problem, numerous researchers have conducted studies in regard to extrinsic tactile sensing. Among these, optical tactile sensors are proposed and focused, which utilize cameras within the fingertip to capture the image of the deformed soft surface after contact, reconstruct the shape, and estimate the forces[6, 7]. Chorley et al.[8] utilized a thin rubber skin with markers to emulate human fingertips biologically and explored the image of contacting distinct objects. Do et al.[9] applied a hemispherical soft gel fingertip and leveraged transfer learning to estimate force precisely under large deformation. Fernandez et al.[10] introduced a flexure into the fingertip, realizing compliance and measuring forces in the meantime. Despite detailed shape reconstruction, forces and torques are difficult to acquire straightforwardly, often based on neural networks to estimate. Besides, the structure and system are complicated where cameras and illumination are required.

Tactile array sensor obtains forces and torques induced by contact more intuitively[11, 12]. Beebe et al.[13] explored a silicon-based single-point sensing element to perceive force and loading direction. Wettels et al.[14] extended multiple electrodes

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in the fingertip to evaluate contact force distribution. Lin et al.[15] developed a piezoelectric tactile sensor array with a multilayer structure that distinguishes different stimulus modes. To increase array density, Oh et al.[16] realized a 100 μm-sized sensor unit and used the array to record normal and shear forces with high resolution. Researchers strive further to integrate sensor arrays into curved surfaces. Ying et al.[17] presented a fabrication method for integrating electronics onto closed-tube surfaces. Kõiva et al.[18] applied laser structuring technology to manufacture 3D-shaped tactile sensors. Weichart et al.[19] wrapped capacitive sensing strips around curved soft fingertips. When focusing on devising dense sensor arrays, challenges emerge in complex and expensive manufacturing technologies, low sampling frequency, and extensive electronic systems[20–22], especially when precise shape reconstruction is not pursued. Consequently, several studies have endeavored to integrate sensing and the compliance behavior of the fingertip, achieving tactile sensing and pose estimation in the meantime. Yu et al. [23] settled a pyramid bump above electrode arrays, which react accordingly to the deformation of the bump caused by distinct forces. Boutry et al.[24] devised microstructures of different scales between the capacitor arrays, which deform and respond diversely to distinct external forces. Ham et al.[25] developed a porous dielectric structure that produces and propagates deformation to the sensors on its sides, capable of monitoring normal, shear, and torsional stresses. To sum up, sensor-integrated compliant fingertips with a miniaturized electronic system are highly desirable for high-frequency and high-precision multiaxial force and torque sensing.

To this end, this paper aims to propose a compact, low-cost, and easy-to-fabricate compliant fingertip for daily scenarios requiring multi-dimensional force and torque sensing, as well as pose estimation. A curved flexible beam array is utilized to construct the fingertip, whose layout provides different deformation for each beam under distinct exerted wrenches. On this basis, a strain gauge sensor based on Flexible Printed Circuit (FPC) technology is designed to obtain local deflection, which possesses low hysteresis, long-term durability, and high repeatability. Besides, a compact sampling system is devised with wireless transmission at over 1 kHz, guaranteeing real-time sensing. Several validation experiments are carried out to exhibit the sensing capacity and resolution of normal and shear force, bending and twisting torque, and the accompanied pose. Eventually, the compliant fingertip is integrated into the commercial OnRobot gripper and is demonstrated in a wine glass arrangement operation for various processes, showing its potential for daily applications.

This paper is organized as follows: In Sec. 2, the compliant fingertip design and sensor fabrication and characterization are presented. Section 3 provides the validation experiments of force and torque sensing and the corresponding pose estimation. Besides, the potential daily application of arranging wine glasses is presented. Finally, conclusions are drawn in Sec. 4.

2. SENSOR-INTEGRATED COMPLIANT FINGERTIP DESIGN

2.1 Mechanical Design based on Flexible Beam Arrays

The multi-dimensional force and torque sensing capability of the compliant fingertip is achieved by integrating local information from multiple contact regions. As demonstrated in Fig. 1(a), the compliant fingertip is constructed based on a 2×2 array of contact regions. In each region, one flexible beam is utilized as the contact and sensory unit. The flexible beam is devised as a convex curved segment with rectangular cross sections of the same size, similar to the structure of human fingertips. The width of the beam is 9.8 mm and the total curve length is 31.12 mm. The beam is constrained at both ends. A revolute pair is applied at one end, while a prismatic pair is at the other. Both the revolute and prismatic pair are attached to the finger base. To provide compliance, the beam is manufactured based on 3D printing using PLA material. As a consequence, deflection of the beam occurs under contact with external forces and torques are exerted. The thickness of the beam is determined as 0.6 mm, capable of dealing with forces at levels of ten newtons. The height of the beam, relative to the revolute pair, is 9 mm while the largest displacement of the prismatic pair is 8 mm, which provides sufficient space for deflection when contacting and manipulating objects.

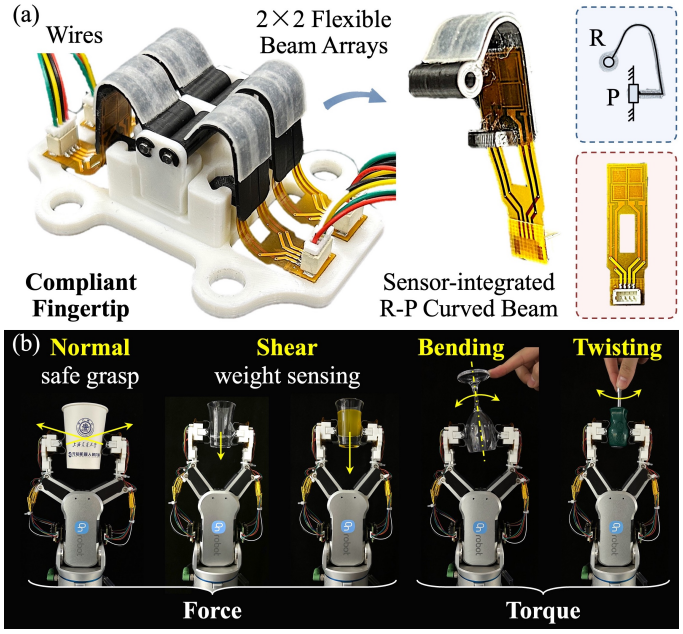


FIGURE 1: OVERALL DESIGN OF THE COMPLIANT FINGERTIP WITH SENSOR-BASED FLEXIBLE BEAM ARRAYS. (A) STRUCTURAL DESIGN AND SENSOR DETAILS. (B) SENSING CAPACITY.

The flexible beams are arranged in a 2×2 array, with the revolute pairs fixed to the middle of the finger base and the prismatic pairs to the outer sides. On one side of the finger base, the revolute pair of the two flexible beams are coaxial. Meanwhile, the interval between the two flexible beams on the same side is 2 mm. In this case, the overall size of the compliant fingertip with four flexible beams is $30 \times 25 \times 25\, mm^3$. Under unloaded conditions, the contacting area, constructed by the highest points of the beams, is of the size $22 \times 25\, mm^2$. In addition, a piece of silica gel with 0.4 mm thickness and 20 mm length is attached to the contact surface of the flexible beam to enhance the friction coefficient. As shown in Fig. 1(b), the proposed compliant fingertip is capable of sensing normal and shear forces, as well as bending and twisting torques. In addition, the compliant fingertip is easy to integrate into existing commercial grippers, such as

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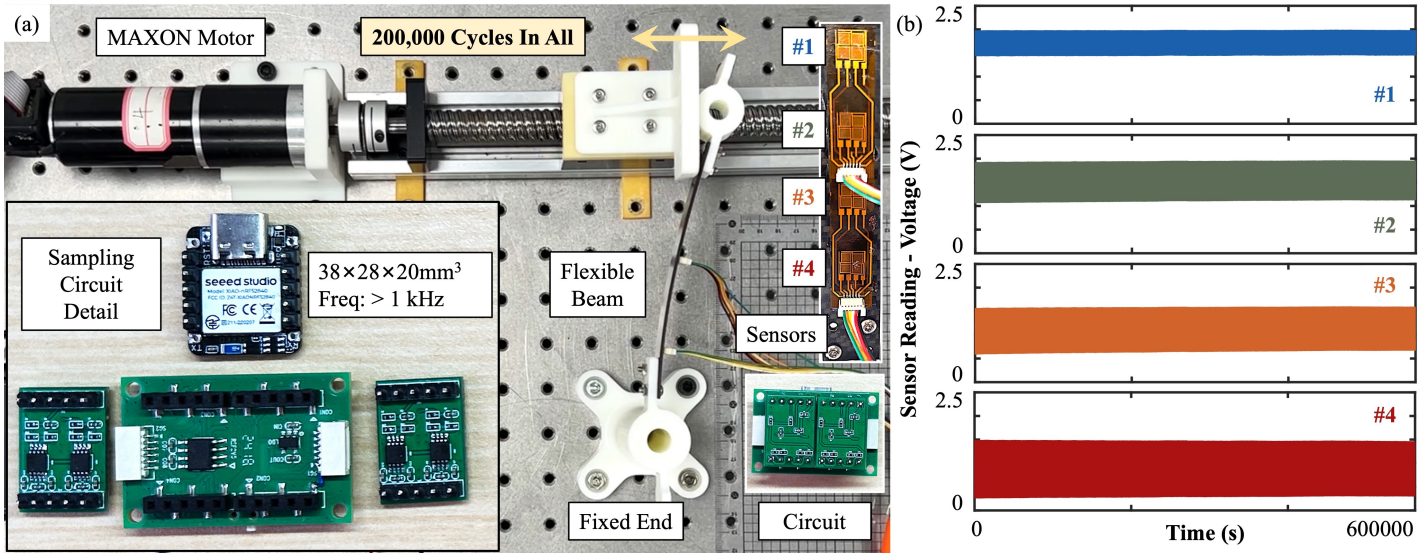


FIGURE 2: (A) CHARACTERIZATION EXPERIMENT SETUP AND (B) ENTIRE SENSING READINGS FOR THE FPC-BASED SENSOR.

OnRobot, to grasp and manipulate household supplies, laying the foundation for human-robot interaction in daily scenarios.

2.2 Design and Fabrication of Strain Gauge Sensor

Compared with sensors using photo-etch technology, FPC-based sensors are much cheaper and easier to manufacture. Therefore, a strain gauge sensor with a Wheatstone bridge configuration is devised as shown in Fig. 1(a), which contains a 2×2 grid. Specifically, each two grids in the diagonal line share the same direction of resistance wires. As a consequence, the full-bridge form strain gauge is capable of responding to the local deflection where it is attached to the beam and is resistant to temperature changes. In addition, it can be deduced that the strain gauge reading is linear to the local deflection under the circular curvature hypothesis. To make the sensor easy to fabricate and cost-efficient, the traditional copper is used for etching, which is often used for Printed Circuit Boards (PCB) while the minimum line width is greater than $50\,\mu m$. On this basis, the sensor can be designed according to the requirement with a large freedom, including the number, position and pose of strain gauges in one piece. Moreover, the sensor can conform to curved surfaces readily due to the flexibility of the PI substrate and thus is suitable to be used in compliant fingertips.

2.3 Comprehensive Characterization of Sensor

To characterize the proposed sensor, an experimental validation is carried out. As illustrated in Fig. 2(a), a sensor-integrated straight flexible beam is applied with one end fixed and the other pushed back and forth by a block. The flexible beam is made of carbon fiber with a thickness of $1.5\,mm$ and a length of $140\,mm$. Two FPC-based sensor pieces are attached to the surface of the beam with two strain gauges designed at an interval of $23\,mm$ in one piece. The pushing block is actuated by a MAXON Motor, which produces a cycle at the frequency of $0.3\,Hz$. From the figure, the maximum deformation of the flexible beam is remarkable where the maximum strain at the strain gauge closest to the fixed

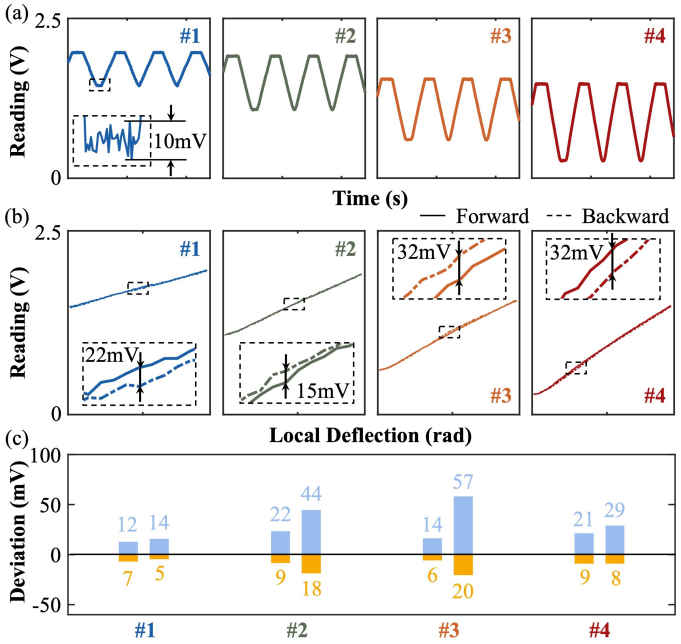


FIGURE 3: CHARACTERIZATION OF THE PROPOSED SENSOR. (A) PARTIAL ENLARGED DETAILS OF SENSING READINGS. (B) HYSTERESIS CHARACTERISTIC CURVE IN ONE CYCLE. (C) DEVIATIONS OF EXTREME VALUES DURING THE ENTIRE LOADING.

end reaches $6000\,\mu\varepsilon$, calculated based on the discretization-based modeling method for flexible rods under large deflection in [26]. It should be noted that the resistance of the strain gauge is small (about $8\,\Omega$ in the size of $10 \times 10\,mm^2$) due to the utilized technology. Therefore, a specific sampling circuit is devised. As shown in Fig. 2(a), the sampling circuit contains several modules, including a $0.12\,V$ voltage supply, an amplification and offset board, and an MCU with wireless data transmission. The offset is indispensable since the sensor can be conformed to curved surfaces, which will induce an initial nonzero value when no deflection is

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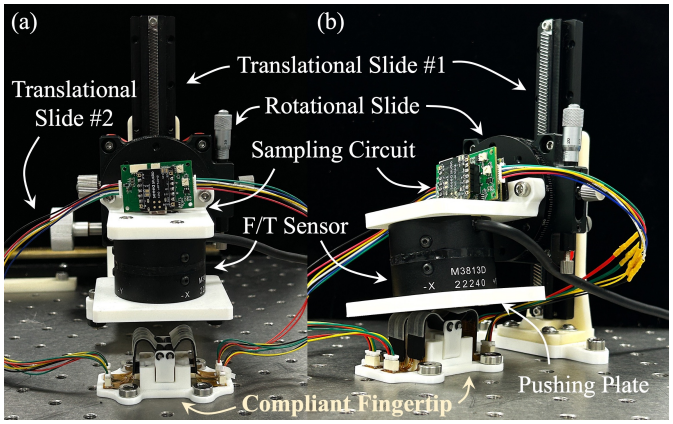


FIGURE 4: EXPERIMENTAL SETUP FOR SENSING VALIDATION. (A) LAYOUT FOR NORMAL FORCE, SHEAR FORCE, AND BENDING TORQUE TESTS. (B) LAYOUT FOR TWISTING TORQUE TESTS.

aroused, reducing the effective range of the sensor. The MCU is not only able to transfer sensor readings through wireless BLE but to sample four strain gauges in series at a highest frequency of over 1 kHz as well. After combination, the overall size of the sampling circuit is $38\times28\times20\,mm^3$, which can be more compact to match with specific sensors. In the experiment, a total number of 200000 cycles is implemented when the sensor readings for the four strain gauges are recorded, as shown in Fig. 2(b), which remain relatively steady for the entire process.

A comprehensive characterization of the proposed FPC-based sensor is provided in Fig. 3. The detailed sensor readings of three cycles are shown in Fig. 3(a), where the maximum fluctuation of the readings under a given constraint can be obtained as $10\,mV$. Compared with the overall $2500\,mV$ range, the fluctuation accounts for 0.4%, which provides a high precision for the sensor. Moreover, one cycle of the sensor readings corresponding to local deflection is selected and depicted in Fig. 3(b), which is related to the block displacement sensed by the encoder of the motor. The readings of pushing forward and backward processes (expressed by solid and dashed lines, respectively) are compared to exhibit the hysteresis property of the sensor. The maximum value difference under the same local deflection is larger when the variance range of the strain gauge increases. The largest difference is about $32\,mV$, indicating a relatively small and negligible hysteresis for the sensor. Figure 3(c) gathered the readings under the two extreme conditions of loading for the whole 200000 cycles. The left bar correlates to the unloaded case while the right one to the fully loaded case. All the readings are treated to minus the average values, showing the maximum deviation. The largest deviation is $77\,mV$, which is 8.3% with respect to the variance range, indicating that the sensor maintains its performance even after long-term usage. From the variation tendency of the readings, it can be revealed that the values under extreme conditions change gradually, potentially attributing to plastic deformation of the beam. Consequently, the deviation can be considered smaller under the same loading condition, demonstrating a high repeatability for the sensor. From the validation results, the proposed FPC-based sensor exhibits high precision, low hysteresis, high repeatability, and good durability for practical applications.

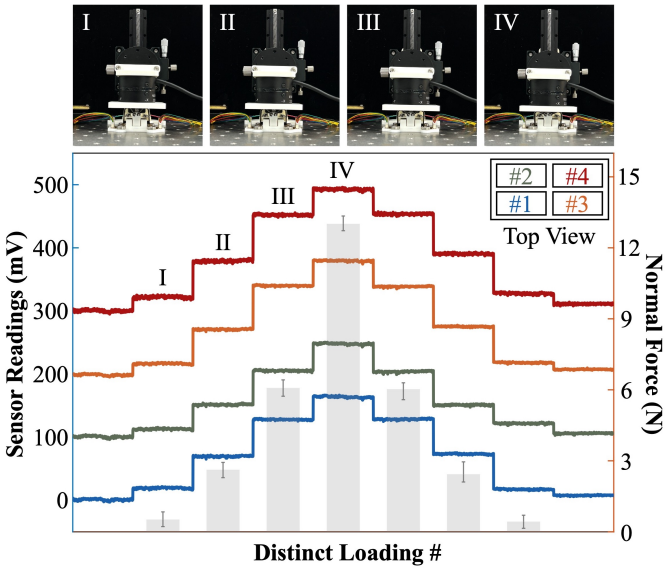


FIGURE 5: NORMAL FORCE SENSING WITHOUT FRICTION.

3. VALIDATION EXPERIMENTS AND APPLICATION

3.1 Experimental Setup

To validate the multi-dimensional force and torque sensing, along with the pose estimation capability, validation experiments are conducted, as illustrated in Fig. 4. The normal force, shear force along the beam length direction, bending torque, and twisting torque are tested. One compliant fingertip is fixed on the experimental table, loaded through a pushing plate. A commercial F/T sensor is connected to the pushing plate to obtain the theoretical value of the forces and torques. The pushing plate is actuated by a translational slide and a rotational slide, providing the required normal force and the bending and twisting torques through altering poses. The loading device can further be fixed to another translational slide, which provides shear force. The accuracy of the translational slide is $0.1\,mm$ while that of the rotational one is 0.02°, which is well precise for the compliant fingertip. The normal force, shear force, and bending torque are exerted through the layout shown in Fig. 4(a), while the twisting torque is applied through the layout shown in Fig. 4(b). The F/T sensor records data at a frequency of $120\,Hz$ and the sampling circuit transmits sensor readings through wireless BLE at $100\,Hz$ to match the F/T sensor, which also leads to a higher precision due to filtering. The results demonstrate the capability and the minimum resolutions of sensing forces and torques, and the pose.

3.2 Sensing of One Compliant Fingertip

Normal force sensing is achieved by moving the translational slide to different positions. Figure 5 demonstrates the variance of the four sensors under 4 distinct loaded conditions, sequentially loading and unloading at an interval of $2\,mm$ for the pushing plate. It should be mentioned that the silica gel provides a non-negligible friction if the loading changes from a previous position to the next. Therefore, the compliant fingertip is pressed and released to contact the plate for each loading position, ensuring that the contact state is unaffected by the loading and unloading processes. The sensor reading for the unloaded state is regulated

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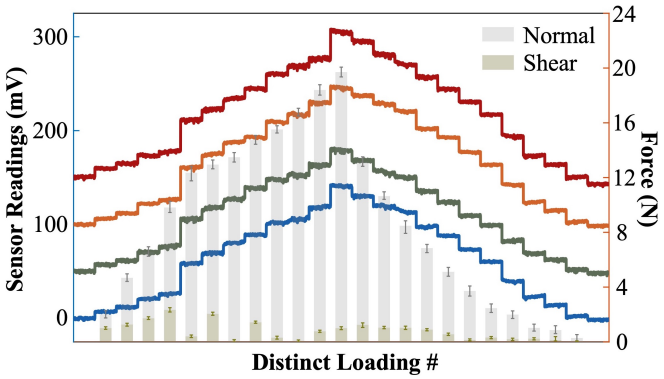


FIGURE 6: NORMAL FORCE SENSING CONSIDERING FRICTION.

to a given value, which is convenient for observing the reading variance when the normal force increases. Since the compliant fingertip undergoes different deformation under distinct loading conditions, the corresponding contact stiffness alters, which results in a variable minimum resolution for the normal force. the normal force increases to 0.5 N from unloaded state to state I, while the sensor readings increase 14 mV, two times the fluctuation, indicating a distinguishable change. As the loading proceeds from state II to IV, the normal force grows to 2 N, 6 N, 13 N. Meanwhile, the sensor reading increases from 70 mV, 130 mV to 160 mV (sensor #1). It can be seen that the growth of the readings shrinks when the normal force is larger, and thus, the resolution is larger. For normal force sensing, minimum resolution appears near the initial loading process, which is smaller than 0.5 N.

During practical applications such as grasping, friction is indispensable and affects the contact state. To evaluate the influence of friction on the normal force sensing, a continuous loading and unloading process is implemented. As demonstrated in Fig. 6, one cycle of 12 distinct loaded conditions is achieved. Similarly, the sensor readings alter accordingly to the normal force and the minimum resolution, smaller than 0.3 N, which corresponds to 0.5 mm, occurs at the last unloading state. However, the readings and the force are no longer symmetric. When the pushing plate descends after contacting the fingertip, the contact point does not slip due to friction, causing the normal force to increase rapidly. The sensor readings experience a sudden elevation at a point where slip occurs. This can be observed through the measured absolute value of the shear force in the figure, corresponding to a great shrinkage. Several slip phenomena happen during the loading process and are related to sensor reading variances. As a result, the influence of friction is reflected by the readings. Take the maximum loading condition, the normal force is 20 N with friction, which is larger than 13 N without friction. Meanwhile, the sensor reading is 140 mV, compared to 160 mV for friction-free case (sensor #1). Moreover, for the same position of the pushing plate, different frictions lead to different contact states and distinct sensor readings, implying the potential of the proposed compliant fingertip to sense shear force.

To further investigate the capability and resolution, 6 distinct loading conditions are applied. As shown in Fig. 7, the compliant fingertip is first pressed downward to state I, which corresponds to a normal force of about 16 N. Friction exists since the beams

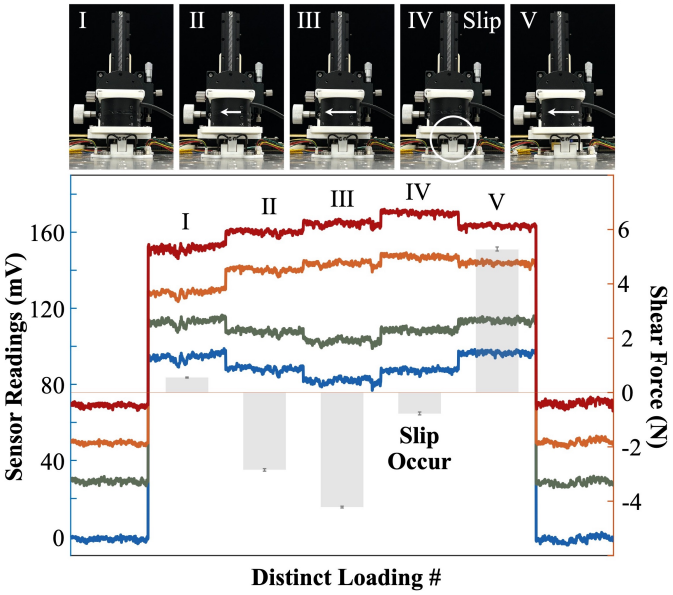


FIGURE 7: SHEAR FORCE SENSING OF THE FINGERTIP.

are not in contact after release. Subsequently, the translational slide moves at an interval of 1 mm to state V. During the loading process, the readings for sensors #1 and #2 share a similar tendency, while those for sensors #3 and #4 share a distinct yet similar trend due to the symmetric layout. Further, the reading variances for these two sets of sensors are opposite from I to III, revealing an increase of shear force. However, all the readings synchronously increase from III to IV, indicating an occurrence of a slip, which can be observed by a sudden shrinkage of shear force. The readings of the two sets alter inversely again from IV to V, corresponding to another shear force growth. As a result, the sensor-integrated fingertip not only possesses shear force sensing capacity but can also detect slip occurrence. In addition, the resolution for shear force differs greatly according to contact conditions. From the involved experiments, the minimum reading variance appears from II to III, which is 3 mV (sensor #1), relating to a shear force increase of 1.38 N. However, this reading increase is undetectable. A more convincing resolution is 3.37 N increase, along with 1 mm, from I to II near the initial loading process with a reading variance of 9 mV. This result is highly dependent on the contact condition of the fingertip and can be further improved by refining the sensor and the sampling circuit.

An exerted pure bending torque leads to the same deformation for one side of the compliant fingertip while distinct for two sides. In this case, bending torque can be sensed through the different variances of the sensor readings. As depicted in Fig. 8, one cycle of 6 distinct loading conditions is applied, with the angle of the pushing plate changing from -10° to 10° at an interval of 4°. The loading process is carried out continuously, which takes friction into consideration to simulate practical operations. The sensors are grouped into two sets, same as the shear force results, in which the readings share a similar variance tendency due to symmetry. The readings reveal that the sensors at the pressed side alter more significantly than the other side when the loading angle is large. In cases such as sensors #1 and #2 from state I

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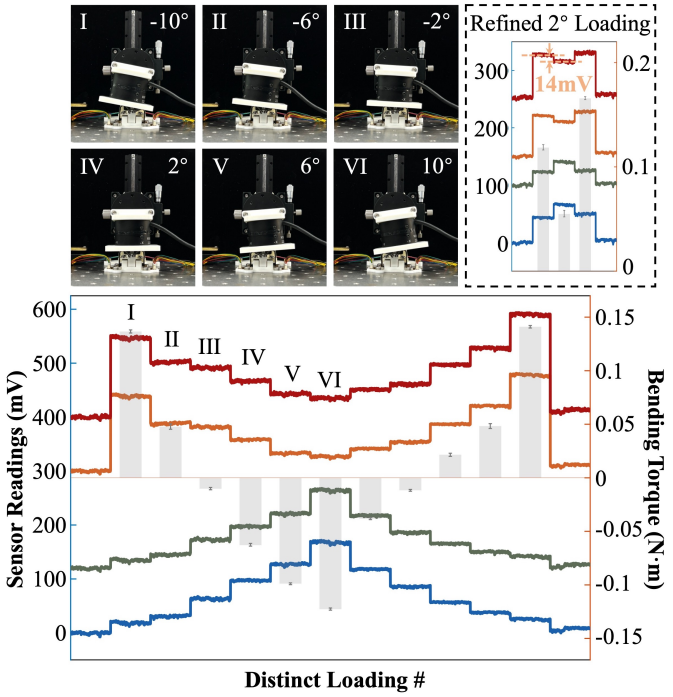


FIGURE 8: BENDING TORQUE SENSING OF THE FINGERTIP.

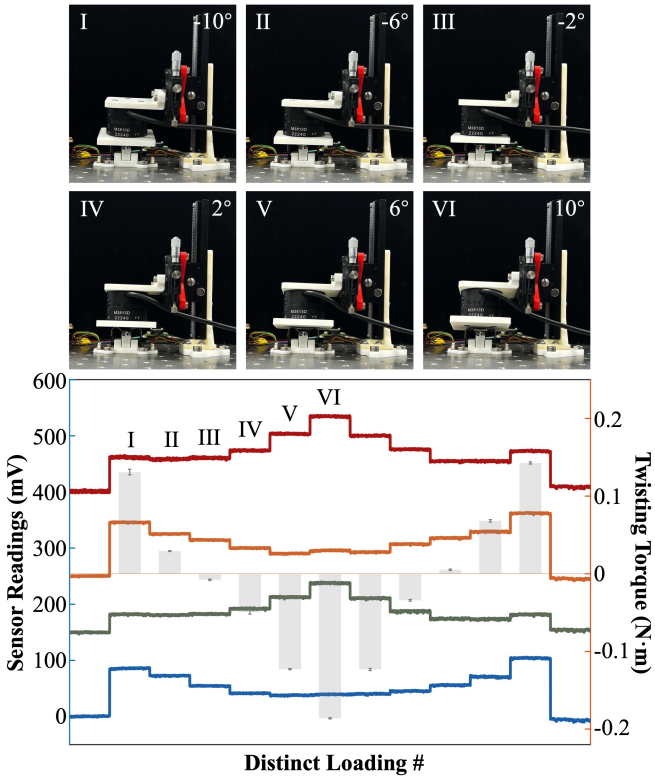


FIGURE 9: TWISTING TORQUE SENSING OF THE FINGERTIP.

to II and sensors #3 and #4 from state V to VI, the reading variance, 7 mV, is small and undetectable. This may be attributed to one side of the flexible beams being detached from contact or only slightly pressed. Although these sensors are unable to distinguish bending torques, those at the pressed side showcase a great change. Therefore, the whole compliant fingertip exhibits a bending torque sensing ability in a wide range of poses. Likewise, the resolution for the bending torque alters according to the contact condition. From the results, it can be calculated that the bending torque resolution is larger at $\pm 10°$ and near $0°$, which are $0.084\,N \cdot m$ for $46\,mV$ variance and $0.052\,N \cdot m$ for $28\,mV$ variance, respectively. A more refined loading process from $0°$ to $-2°$ and back to $0°$ is recorded, where the minimum bending torque resolution appears to be $0.063\,N \cdot m$ for $14\,mV$ variance. In accordance, the rotational angle resolution is $2°$.

The loading equipment is moved into a different position to exert twisting torques. The same continuous loading cycle as the bending torque is utilized. Figure 9 demonstrates the sensor readings, where two sets with each similar variance can be grouped, including sensors #1 and #3 and sensors #2 and #4. Consistently, the readings change greatly for the pressed side and slightly for the other side when the loading angle is large. This phenomenon is more obvious and the critical loading angle is smaller than that in the bending torque experiment, where only $3\,mV$ change from I to III for #2 and #4 and imperceptible $1\,mV$ change from IV to VI for #1. The critical loading angle refers to when one side of the fingertip is out of contact. Moreover, the readings of the pressed side alter less than that for the bending torque under the same loading condition. This is because the 2×2 grid shares a different length and width for the contact regions. When twisting torque is applied, the distance between the contact regions for the two sides is shorter than when bending torque is applied. Therefore, the sensor readings are less sensitive to the torque variance, which leads to a larger resolution. In the meantime, the resolution maintains to be larger at $\pm 10°$ and near $0°$. Specifically, the resolution is $0.063\,N \cdot m$ for $27\,mV$ variance at $10°$. Within the range of critical loading angle, the minimum resolution is $0.042\,N \cdot m$ for $15\,mV$ variance from III to IV, along with a $4°$ rotational angle resolution. It should be noted that the calculated resolution for twisting torque should be theoretically larger than bending torque. The adverse result of torque resolutions is ascribed to different contact states, where the average normal force is about $2.88\,N$ for the twisting torque experiment, smaller than $4.78\,N$ for the bending torque experiment. However, the loading angle resolution is independent of contact states, which is verified to be larger when exerting twisting torque on the fingertip.

3.3 Wine Glass Arrangement Demonstration

The proposed sensor-integrated compliant fingertip is readily adaptable to be mounted onto commercial grippers, replacing existing tip ends. As demonstrated in Fig. 10(a), the OnRobot gripper is equipped with the devised compliant fingertips. For each fingertip, four FPC-based strain gauges are adopted, whose readings are recorded through one sampling circuit and transmitted to the upper processor through wireless BLE. The communication rate is confined to $50\,Hz$, although available at $1\,kHz$ at most, providing sufficiently real-time and precise sensor readings. It should be mentioned that the maximum fluctuation for the sensor readings under stable conditions is about $5\,mV$.

The detailed wine glass arrangement operation can be separated into 9 stages, as illustrated in Fig. 10(b). Meanwhile, the

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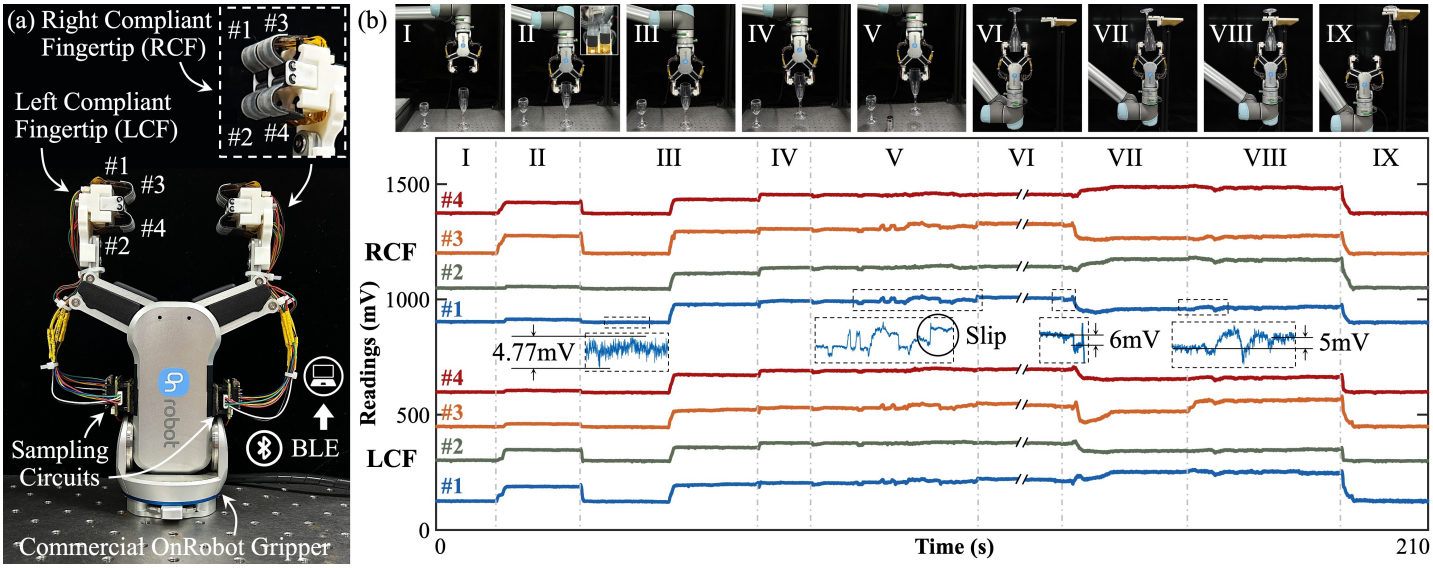


FIGURE 10: WINE GLASS ARRANGMENT DEMONSTRATION BASED ON COMPLIANT FINGERTIP-INTEGRATED GRIPPER. (A) OVERALL FRAMEWORK AND COMMUNICATION FORM OF THE INTEGRATED GRIPPER. (B) STAGES OF GLASS ARRANGEMENT OPERATION WITH SENSING READINGS DURING THE ENTIRE PROCESS.

sensor readings are recorded accordingly, which correspond to the labeling shown in Fig. 10(a). Some detailed reading variances of sensor #1 for the right compliant fingertip (RCF) are supplemented in the figure to emphasize sensing results. In stage I, the gripper is connected to a collaborative robotic arm in the initial position, on top of the wine glass with an unknown location. A 200 g weight is placed in the wine glass. The robotic arm drives the gripper to grasp the wine glass for a first attempt in stage II. From the readings, the sensors #1 and #2 for the left compliant fingertip (LCF) and sensors #3 and #4 for RCF change more greatly than the other four sensors, indicating a poor grasping condition that the gripper fails to grasp the wine glass in the central line. The experimental picture for II verifies the misalignment of the attempt. Subsequently, the gripper opens and the robotic arm adjusts to the proper position for a second attempt during state III. Since the readings for all eight sensors increase at a relatively consistent trend, referring to an effective grasping condition. On this basis, the robotic arm lifts the wine glass up in stage IV. It can be seen that all readings grow slightly from III to IV. However, this is not due to shear force induced by the weight increase. Instead, the OnRobot gripper maintains a settled force control during the whole process, which is responsible for the variance. Stage V is set to detect the shear force. The weight is removed from the wine glass and several times of pushing and pulling are applied to the wine glass. From the results, the 2 N decrease is hard to distinguish with the variance mingled with fluctuation. Nevertheless, the intentionally imposed larger shear force can be detected. It should be further noted that the readings revert to previous values when the first few exertions are released, whereas they change to distinct values for the last time, implying a slip occurrence. In stage VI, the robotic arm flips the wine glass top to bottom and approaches the rack with an unknown location. Stage VII correlates to the first attempt. The readings for sensors #1 and #3 and sensors #2 and #4 of RCF alter in an inverse direction, with a decrease for the former and an increase for the latter. This signifies a bending torque induced to the wine glass, which indicates that the robotic arm fails to match the wine glass with the opening of the rack. In addition, a more detailed variance can be observed at the beginning of VII, where the readings for RCF undergo a 6 mV drop, suggesting that the bumping of the wine glass and the rack first arouses an extrusion towards the LCF and releases the contact force exerted on RCF. For LCF, the readings vary consistently opposite. The readings for sensors #1 and #3 increase while those for sensors #2 and #4 decrease due to the imposed bending torque. Nonetheless, sensor #3 demonstrates an abnormal variance, with its reading falling before rising. This phenomenon that one of the eight sensors alters differently from the expected variance caused by a bending torque is invariably observed during multiple rounds of experiments, which is ascribed to the fact that a twisting torque is introduced during the bumping process. The wall of the wine glass is nearly circular with a diameter of 49 mm, resulting in a non-planar contact area with the fingertips. Moreover, the fingertips are not perfectly aligned with the central line of the wine glass. As a result, the wine glass rotates when it bumps into the rack, releasing one contact region while imposing the others. When rotation is settled, all the readings subsequently change as expected. Likewise, the sensor #3 for LCF is pushed back with an obvious increase in reading when the robotic arm drives the wine glass backward during stage VIII. From the experimental pictures, the wine glass rotates due to the applied bending torque in VII and recovers slightly in VIII, which is revealed by a 5 mV reading increase. Due to lacking enough friction, the wine glass is unable to completely recover to its original vertical state. It should be further mentioned that the reading variances for sensors #2 and #4 are notably smaller than those for sensors #1 and #3, both for LCF and RCF. This is because sensors #2 and #4 are closer to the gripper, causing a deeper impact due to the continuous force control of the OnRobot

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gripper. Eventually, in stage IX, the robotic arm moves and releases the wine glass in accordance with the opening of the rack to accomplish the arrangement.

3.4 Discussion

The proposed compliant fingertip demonstrates the capacity of force and torque sensing, along with pose estimation, which exhibits potential in usage for various applications. However, there are still limitations and space for improvements. It can be observed from the sensing readings in the validation experiments that the data at the beginning and at the ending of the experiments deviate from each other a little. This is in part attributed to the material and manufacturing technology of the flexible beams since the utilized 3D printing technology leads to a slight deviation when constructing different units with the same designed curve. Therefore, calibration should be carried out before applying the compliant fingertip to overcome the inconsistency. Several improvements can be achieved to ameliorate the shape precision and thus, increasing sensing accuracy. One direct way is to use metallic materials, such as spring steel (typically 65Mn), beryllium bronze. Mature manufacturing technology can ensure the accuracy of the beam shape. Moreover, the validation experiments are focused on demonstrating the capacity of force and torque sensing, as well as the pose. The sensing readings are recorded with accordance with force, torque, and pose. Nevertheless, the estimation of these variables based on the sensing readings require reverse experiments and calibration, which will be considered in future studies. It should be noted the theoretical modeling of the flexible beams can be accomplished through the discretization-based modeling method proposed in [26]. In this case, a more comprehensive force and torque resolution and pose reconstruction can be developed and analyzed at the same time, which enlarges the potential of practical usage in various industrial and daily applications.

In addition, future studies will be focused on improving the sensing performance. On one hand, the sampling circuit should be further integrated into the fingertip design for a more compact form. On the other hand, the shape of flexible beams will be further optimized. It should be mentioned that different initial curves of the beams lead to distinct stiffness characteristics under contact. As a result, optimal curve should be obtained for required fingertip properties. Besides, the array arrangement can also be optimized. The proposed 2×2 array provides a compact form and a basic distinguishing capacity for force and torque, where 3×3 or other formats may showcases a better sensing performance. Further, sensing-based control can be achieved for compliant fingertips to operate within real-time scenarios.

4. CONCLUSION

In this paper, a compliant fingertip is presented, constructed by a 2×2 grid of flexible beam arrays that integrate sensors. The devised FPC-based strain gauge sensor with the sampling circuit provides high-precision deformation information wirelessly with low hysteresis, high repeatability, and good durability at over $1\,kHz$ frequency. Through validation experiments, the proposed compliant fingertip possesses the sensing capacity for multi-dimensional force and torque along with pose estimation. Finally, a wine glass arrangement demonstration is settled based on the fingertip-integrated commercial gripper, showcasing its sensing performance and potential usage in practical applications.

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